# Mixed First-Detection Patterns and Geographic Concentration of Sequenced Hospital-Associated SARS-CoV-2 Cases in Hong Kong, May–August 2022: A Secondary Analysis

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Secondary analysis of Gu et al. (2026)

*Revised draft after peer-review reframing — not a claim of dual seeding or incidence excess*

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**Abstract**

**Background.** Gu et al. characterised hospital-associated SARS-CoV-2 clusters in Hong Kong during a period of relaxed visitation (28 May–18 August 2022), confirming 18 nosocomial clusters (126 genomes) among 29 investigated ward units (162 genomes) and finding no independent association between community mobility and community-to-hospital introductions after time adjustment [1]. Here we ask two narrower secondary questions: (i) how first-detection order by case type is distributed across confirmed wards, and (ii) whether the geographic distribution of the sequenced sample persists under crude health-system throughput scaling, once genomic representation is audited against official reporting.

**Methods.** We reanalysed the published anonymised plot table. Primary analyses used the 18 parent-confirmed wards ($n = 126$); sensitivity analyses used all 29 investigated units ($n = 162$). Ward-level first-detection order compared earliest staff/outpatient versus inpatient detection dates (same-day and $\pm$1–2-day windows as sensitivity). Inpatient counts were scaled by HA beds, discharges, and patient-days; staff/outpatient counts were reported separately. Unrepresented HA clusters were coded as missing genomic representation, not zero burden. Leave-one-hospital, leave-one-ward, and BA.5.6-ward exclusions assessed robustness.

**Results.** Among confirmed wards, both staff/outpatient-before-inpatient (4 wards, 38 persons) and inpatient-before-staff/outpatient (5 wards, 48 persons) first-detection orders were observed, plus same-day and single-type patterns. A regional Fisher comparison of ordered pathways was severely underpowered (OR 0.75, exact $p = 1.0$, 95% CI 0.03–17.5; $n = 9$ wards). NTWC+NTEC held 103/126 (81.7%) sequenced confirmed-ward persons and 68/87 (78.2%) sequenced confirmed-ward inpatients. Pooled NT/urban ratios of sequenced inpatients were approximately 4.0$\times$ per 1,000 beds, 4.9$\times$ per 100,000 discharges, and 4.1$\times$ per 100,000 patient-days among represented clusters; leave-one-out pooled bed ratios ranged $\sim$2.6–5.5, and excluding the BA.5.6 ward reduced the pooled bed ratio to $\sim$2.8. Official bulletins mentioned hospitals in clusters absent from the genomic table (notably KCC and HKWC), so those clusters cannot be treated as zero burden.

**Conclusions.** The sequenced sample shows mixed first-detection patterns and New Territories concentration after crude inpatient-throughput scaling. Geographically incomplete and potentially differential genomic ascertainment prevents inference about underlying nosocomial incidence, density-driven risk, psychiatric causation, or visitation policy effects. Complete hospital-level outbreak, staffing, and sequencing denominators are required for selection-aware analysis.

**Keywords:** SARS-CoV-2; hospital-associated infection; genomic surveillance; first-detection order; ascertainment; Hong Kong; secondary analysis

# 1 Introduction

Gu et al. sequenced 162 genomes linked to 29 suspected ward clusters across 17 public hospitals (28 May–18 August 2022), confirmed 18 nosocomial clusters (126 genomes covering 51.9% of officially reported nosocomial cases in the window), described predominantly intraward transmission, and found no significant mobility association with introduction rates after adjusting for time trends [1]. Phylogenetically supported staff-before-inpatient order was reported in only 2/18 confirmed clusters [1].

International work already shows that hospital outbreaks may involve patients, healthcare workers (HCWs), visitors, or multiple introductions; that staff–patient networks transmit in both directions; and that epidemiologically labelled outbreaks may hide several genomic introductions [2–6]. Psychiatric and long-stay settings pose distinctive IPC challenges [9–11], and facility outbreak geography is not reducible to urban density alone [12, 13]. Hong Kong hospitals faced intense Omicron pressure despite high HCW vaccination coverage [7], and multi-bedded cubicles facilitated onward spread [8].

**What this note adds.** This is a post hoc secondary analysis of the Gu et al. plot table. It does *not* re-estimate phylogenies, does *not* identify seeding mechanisms, and does *not* estimate true nosocomial incidence. Its defensible contribution is the combination of: (i) a descriptive ward-level first-detection-order taxonomy across confirmed clusters; (ii) HA-cluster concentration of the sequenced sample under inpatient-compatible throughput denominators; and (iii) an explicit audit of geographic genomic representation against independent official bulletin presence.

**Research questions.**

1. How is first-detection order by case type distributed across parent-confirmed wards?
2. Does New Territories concentration of sequenced confirmed-ward cases persist under crude beds/discharges/patient-days scaling, and what does an ascertainment presence audit imply for interpreting that concentration?

# 2 Parent study versus this note

Table 1 separates inherited findings from new analyses.

Table 1: Contribution matrix: Gu et al. (2026) versus this secondary note.

| Item | Gu et al. 2026 | This note |
|---|---|---|
| 162 / 29 suspected / 18 confirmed / 126 genomes; 51.9% coverage | Primary | Cohort definition only |
| Phylogenetic staff-before-inpatient in 2/18 clusters; mobility GAM null; API / lineage structure | Primary | Cited; not re-estimated |
| Ward-level first-detection-order taxonomy | — | New (descriptive) |
| Inpatient-only throughput scaling of sample concentration | — | New (not incidence) |
| Official-bulletin presence vs genomic representation | — | New audit |
| Leave-one-out / BA.5.6-ward / tie-window checks | — | New robustness |

# 3 Methods

## 3.1 Data

Anonymised plot table from Gu et al. ($n = 162$ sequenced persons; hospitals A–Q; HA cluster; ward; case type; lineage; date) [1]. Wards without a trailing asterisk match the parent-confirmed nosocomial set (18 wards, 126 genomes); starred wards (11 units, 36 genomes) were investigated but not confirmed as nosocomial genomic clusters and are reserved for sensitivity analyses. HA beds (31 Mar 2022) and FY2021–22 discharges and patient-days come from HA Statistical Report extracts [14]. Catchment demography uses 2017 planning projections with core district land areas [15, 16]; these are coarse administrative measures, not closed cohorts. Official HA nosocomial press bulletins supply a presence footprint (hospitals/clusters mentioned), not cumulative case totals (Table S3 was not recoverable here).

## 3.2 First-detection order

For each hospital–ward unit we recorded the earliest detection date among staff/outpatient versus inpatient case types. Categories: staff/outpatient before inpatient; inpatient before staff/outpatient; same-day first detection; staff/outpatient only; inpatient only. This is a *temporal first-detection* descriptor, not infection order, introduction source, or transmission direction. Visitors are unobserved. Staff/outpatient remains a combined category because the plot table does not separate them. Sensitivity: treat detections within $\pm 1$ or $\pm 2$ days as the same window. Regional comparison of ordered pathways used Fisher's exact test with an exact odds-ratio confidence interval; with few wards this test is exploratory and severely imprecise.

### 3.3 Throughput scaling and estimands

We report four distinct quantities:

1. Sequenced confirmed-ward **inpatients** per 1,000 beds, per 100,000 discharges, and per 100,000 patient-days (patient-compatible denominators).
2. Sequenced confirmed-ward **staff/outpatient** counts (descriptive; staff FTE denominators unavailable).
3. All linked sequenced persons per confirmed ward (outbreak size; descriptive).
4. Genomic representation by HA cluster (present vs not represented).

We do *not* call total persons per patient-days an HAI incidence rate. Annual FY2021–22 throughput is an imperfect proxy for an 83-day window; we emphasise relative NT/urban ratios among represented clusters. Pooled ratios (sum of numerators / sum of denominators) and means of cluster-level rates are both reported. Leave-one-hospital-out, leave-one-ward-out, and exclusion of the single BA.5.6 ward assess point influence.

### 3.4 Ascertainment audit

Because hospital-level official totals and sequencing-submission denominators are unavailable, we contrast **presence**: HA clusters appearing in official nosocomial bulletins versus those appearing in the sequenced confirmed-ward table, anchored by the published 51.9% overall coverage statistic [1]. Overall coverage does not bound cluster-specific sampling fractions; differential ascertainment could generate large apparent regional ratios.

### 3.5 Ethics and anonymisation

Analyses use aggregated anonymised hospital codes as in Gu et al. [1]. We avoid publishing inferred code-to-name crosswalks. HA cluster labels provide geographic context already used publicly.

## 4 Results

### 4.1 Mixed first-detection patterns among confirmed wards

Among 18 confirmed wards, ordered dual-type pathways contributed 38 persons in staff/outpatient-before-inpatient wards ($n = 4$) and 48 in inpatient-before-staff/outpatient wards ($n = 5$); same-day first detection added 24 persons in 3 wards; inpatient-only wards contributed 16 (Table 2; Figure 1). No confirmed ward was staff/outpatient-only (those units were starred/non-confirmed). Expanding the same-window definition to $\pm 1$ or $\pm 2$ days did not reassign the ordered pathways in this table (same-day wards remained the only ties).

Among confirmed wards with a clear order ($n = 9$), the NT versus urban $2 \times 2$ counts were 3:4 versus 1:1 (staff/outpatient-before : inpatient-before). Fisher OR $= 0.75$, exact $p = 1.0$, 95% CI 0.03–

17.5. The small sample provided no precise evidence about regional differences in first-detection order; failure to detect a difference is not evidence of equivalence.

Cross-tabulation against confirmation status (Table 3) shows that staff/outpatient-only pathways were confined to non-confirmed (starred) units, while same-day detections occurred only among confirmed wards in this sample.

Table 2: First-detection order among 18 parent-confirmed wards (primary cohort).

| First-detection pattern | Wards | Persons | Inpatients |
|---|---|---|---|
| Staff/outpatient before inpatient | 4 | 38 | 18 |
| Inpatient before staff/outpatient | 5 | 48 | 40 |
| Same-day first detection | 3 | 24 | 13 |
| Inpatient only | 6 | 16 | 16 |
| Staff/outpatient only | 0 | 0 | 0 |

Table 3: First-detection order by parent confirmation status (all 29 investigated units).

| Pattern | Confirmed wards | Confirmed persons | Non-confirmed wards | Non-confirmed pe |
|---|---|---|---|---|
| Staff/outpatient before inpatient | 4 | 38 | 2 | |
| Inpatient before staff/outpatient | 5 | 48 | 2 | |
| Same day | 3 | 24 | 0 | |
| Staff/outpatient only | 0 | 0 | 5 | |
| Inpatient only | 6 | 16 | 2 | |

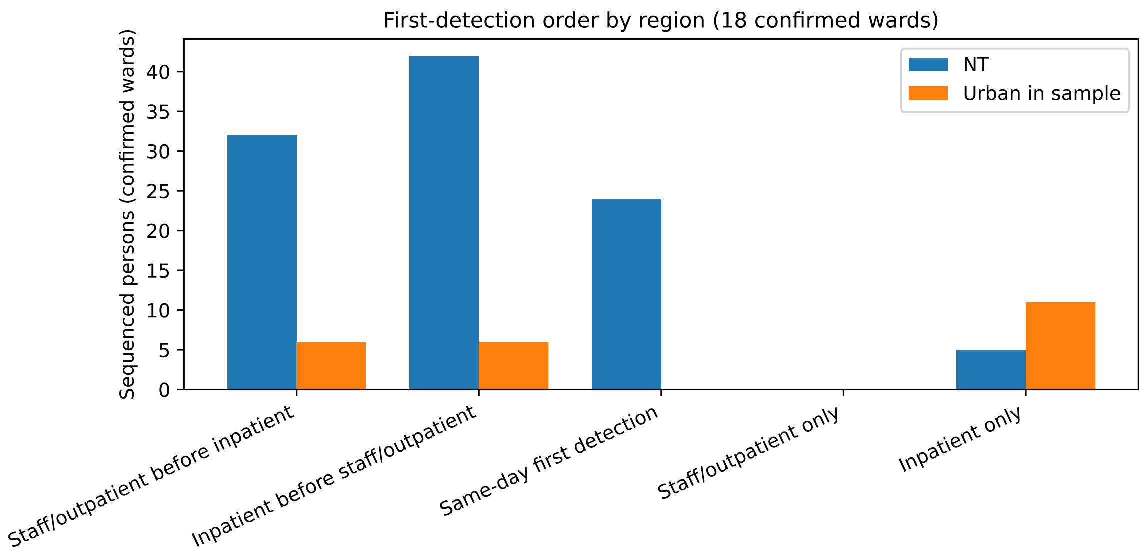


Figure 1: First-detection order composition by region among parent-confirmed wards (case counts). This describes detection timing, not seeding mechanism.

### 4.2 Sample concentration after inpatient-throughput scaling

NTWC+NTEC accounted for 103/126 (81.7%) sequenced persons in confirmed wards and 68/87 (78.2%) sequenced confirmed-ward inpatients (Table 4). Among represented clusters, pooled NT/urban ratios of sequenced inpatients were ∼4.0× per 1,000 beds, ∼4.9× per 100,000 discharges, and ∼4.1× per 100,000 patient-days; means of cluster rates were similar (Figure 2). Leave-one-hospital-out pooled bed ratios ranged ∼2.6–5.5 (median ∼3.9); leave-one-ward-out ranged ∼2.8–5.5 (median ∼3.9). Excluding the BA.5.6 ward reduced the pooled bed ratio to ∼2.8, indicating point influence from a single deep outbreak without erasing NT concentration.

Staff/outpatient sequenced persons (39/126) are reported in Table 4 but are *not* divided by patient-days. KEC appears as a descriptive comparator: relatively low hospital geographic density with low sampled inpatient counts—not a formal negative control.

Table 4: Sequenced persons in parent-confirmed wards by HA cluster. Rates use *inpatient* numerators only. KCC and HKWC are not represented in the genomic sample (NA), not zero burden.

| Cluster | All | Inpatients | Staff/OP | Beds | /1k beds (IP) | /100k disc. (IP) | Representation |
|---|---|---|---|---|---|---|---|
| NTWC | 64 | 44 | 20 | 4,762 | 9.24 | 31.34 | Represented |
| NTEC | 39 | 24 | 15 | 5,144 | 4.67 | 14.69 | Represented |
| KWC | 12 | 8 | 4 | 4,953 | 1.62 | 3.98 | Represented |
| HKEC | 6 | 6 | 0 | 3,307 | 1.81 | 5.87 | Represented |
| KEC | 5 | 5 | 0 | 2,922 | 1.71 | 4.35 | Represented |
| KCC | — | — | — | 6,005 | NA | NA | Not in genomic sample |
| HKWC | — | — | — | 3,076 | NA | NA | Not in genomic sample |

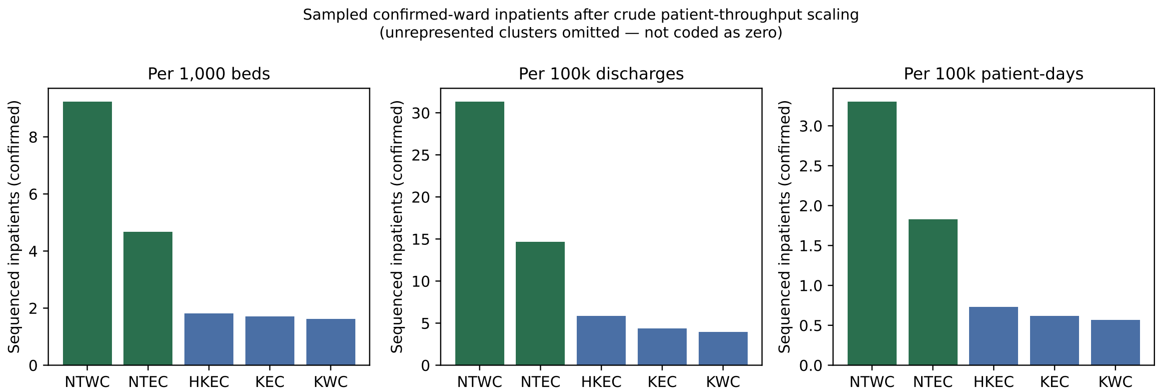


Figure 2: Sequenced confirmed-ward inpatients after crude patient-throughput scaling. Unrepresented clusters are omitted (not coded as zero).

### 4.3 Discordance with coarse density measures

Mean catchment population density and hospitals per 100 km$^2$ were lower for NT than for urban represented clusters, while sampled inpatient rates were higher (Figure 3). We describe this as discordance between *sampled case concentration* and coarse catchment-density measures—not as a proven density paradox of nosocomial incidence. Catchments are administrative; specialty hospitals may draw beyond district borders; staff exposure relates to where staff live and travel; and land-area definitions (including Islands/Lantau handling) can inflate urban density [15, 16].

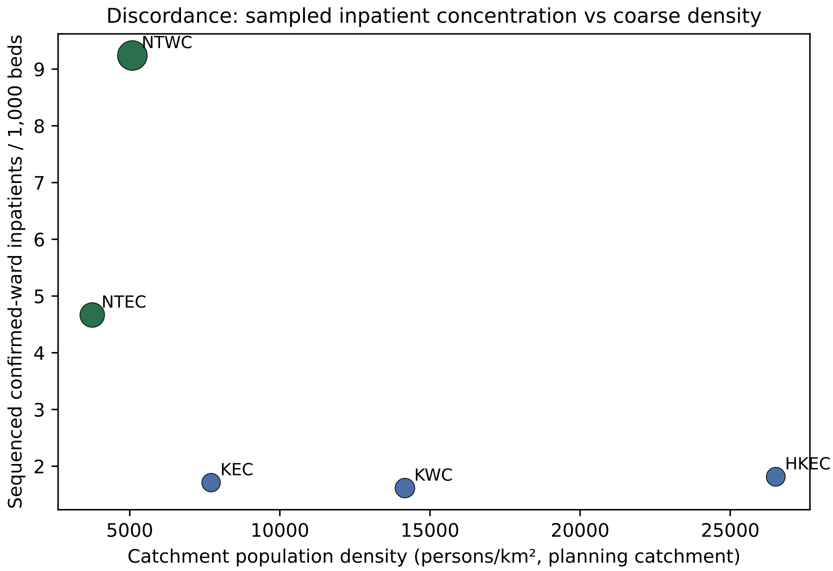


Figure 3: Discordance between sampled confirmed-ward inpatient concentration and coarse catchment population density (represented clusters only).

### 4.4 Geographic genomic representation is incomplete

Public HA bulletins during the window mentioned hospitals across all seven HA clusters, including Kowloon Central and Hong Kong West, which contribute zero confirmed-ward genomes to the plot table (Figure 4). That is missing genomic representation, not evidence of zero nosocomial burden. Parent-study confirmed clusters covered 51.9% of official nosocomial cases overall [1]; cluster-specific sampling fractions remain unknown. Therefore an observed several-fold NT/urban ratio in the genomic table could, in principle, be generated partly or wholly by differential selection into sequencing.

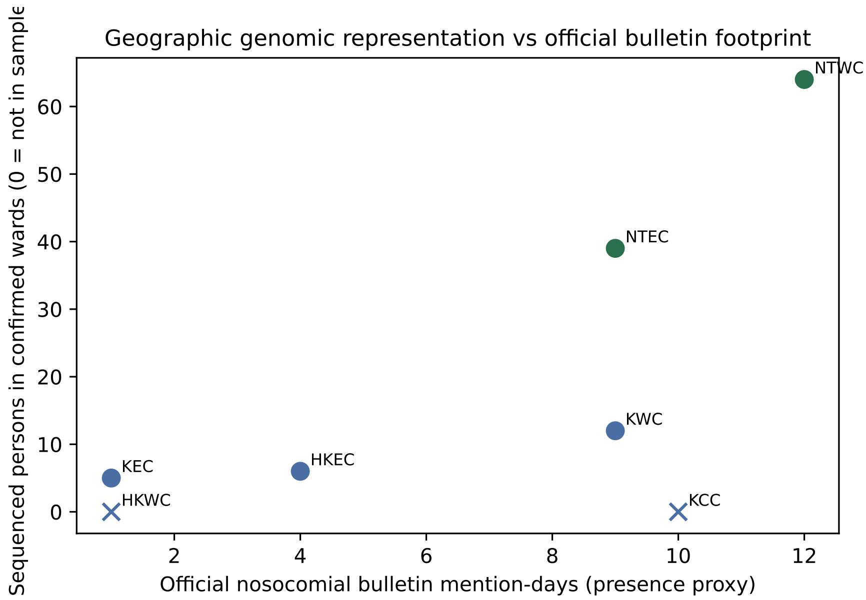


Figure 4: Official nosocomial bulletin mention-days versus sequenced confirmed-ward persons. Clusters on the $x$-axis with zero sequenced persons illustrate incomplete genomic representation.

# 5 Discussion

## 5.1 First-detection order is not dual seeding

Observing both staff/outpatient-before-inpatient and inpatient-before-staff/outpatient detection patterns is consistent with—but much weaker than—international evidence of multi-route hospital transmission [2, 3, 6]. Earliest positive date is not earliest infection, introduction source, or transmission direction. Testing intensity, symptom onset, incubating admissions, unobserved visitors, and outbreak-triggered mass testing can all scramble detection order. The parent phylogenetic staff-index result (2/18) remains the stronger directional claim within this dataset [1]; our taxonomy is complementary and deliberately descriptive.

## 5.2 Sample concentration is not incidence excess

Inpatient-compatible throughput scaling leaves a several-fold NT concentration in the genomic sample. That finding survives crude leave-one-out checks but shrinks when the BA.5.6 ward is excluded, and it does not adjust for $P(\text{detected}) \times P(\text{sampled}) \times P(\text{sequenced}) \times P(\text{included})$. Without cluster-specific official counts and sequencing denominators, we cannot reject the hypothesis that differential ascertainment explains much of the apparent ratio. Psychiatric/long-stay case-mix remains a *candidate* explanation motivated by external literature [9–11] and by NTWC's high mentally-ill patient-day share, but ecological association at cluster scale—especially with one deep ward outbreak—does not demonstrate causation.

### 5.3 Surveillance implication

The geographic distribution of a genomic hospital-outbreak dataset is a joint product of transmission, case mix, outbreak detection, and sequencing selection—not simply underlying disease incidence. That principle is the note's main interpretive contribution. Making it a formal methodological contribution would require selection models, sampling fractions, and preferably validation in another period or system.

### 5.4 Limitations

1. Sequenced confirmed-ward cases $\neq$ complete nosocomial incidence.
2. Annual patient throughput approximates an 83-day window; staff denominators are missing.
3. First-detection order $\neq$ phylogenetic index or true seeding.
4. Bulletin mention-days are a presence proxy, not coverage fractions.
5. Catchment density measures are coarse and partly inconsistent across sources.
6. Regional pathway comparison is severely underpowered.
7. This secondary analysis overlaps the Gu et al. dataset; novelty is limited to the analyses in Table 1.

## 6 Conclusion

Among sequenced persons linked to parent-confirmed hospital clusters, both staff/outpatient-first and inpatient-first detection patterns were observed. Sampled cases were concentrated in New Territories clusters after crude patient-throughput scaling, but geographically incomplete and potentially differential genomic ascertainment prevents inference about underlying nosocomial incidence. Complete hospital-level outbreak, throughput, staffing, and sequencing denominators are required to test whether long-stay or psychiatric case mix—or selection alone—explains the observed concentration.

## Data and code

Analysis scripts: `scripts/6.stage1_peer_review_revision.py` (primary Stage-1 outputs), with earlier exploratory scripts retained for provenance. Tabulated outputs: `analysis/outputs/stage1_*`. Parent genomic methods: Gu et al. [1].

## Competing interests

None declared.

## Acknowledgements

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## References

[1] Gu H., Li M., Wang R., et al. (2026). Genomic epidemiology of hospital-associated SARS-CoV-2 clusters in Hong Kong during a period of relaxed visitation (May–August 2022). *Influenza and Other Respiratory Viruses*, 20, e70249. doi:10.1111/irv.70249

[2] Lindsey B.B., Villabona-Arenas C.J., Campbell F., et al. (2022). Characterising within-hospital SARS-CoV-2 transmission events using epidemiological and viral genomic data across two pandemic waves. *Nature Communications*, 13, 671. doi:10.1038/s41467-022-28291-y

[3] Ellingford J.M., George R., McDermott J.H., et al. (2021). Genomic and healthcare dynamics of nosocomial SARS-CoV-2 transmission. *eLife*, 10, e65453. doi:10.7554/eLife.65453

[4] Benoit P., et al. (including Longtin Y.) (2023). On-demand, hospital-based SARS-CoV-2 genomic epidemiology to support nosocomial outbreak investigations: a prospective molecular epidemiology study. *Antimicrobial Stewardship & Healthcare Epidemiology*, 3, e45. doi:10.1017/ash.2023.119

[5] Haanappel C.P., Oude Munnink B.B., Sikkema R.S., et al. (2023). Combining epidemiological data and whole genome sequencing to understand SARS-CoV-2 transmission dynamics in a large tertiary care hospital during the first COVID-19 wave in The Netherlands focusing on healthcare workers. *Antimicrobial Resistance & Infection Control*, 12, 46. doi:10.1186/s13756-023-01247-7

[6] Abbas M., Robalo Nunes T., Martischang R., et al. (2021). Nosocomial transmission and outbreaks of coronavirus disease 2019: the need to protect both patients and healthcare workers. *Antimicrobial Resistance & Infection Control*, 10, 7. doi:10.1186/s13756-020-00875-7

[7] Wong S.C., Chan V.W., Yuen L.L., et al. (2023). Infection of healthcare workers despite a high vaccination rate during the fifth wave of COVID-19 due to Omicron variant in Hong Kong. *Infection Prevention in Practice*, 5, 100261. doi:10.1016/j.infpip.2022.100261

[8] Wee L.E., Conceicao E.P., Aung M.K., et al. (2023). Nosocomial SARS-CoV-2 transmission in multi-bedded hospital cubicles over successive pandemic waves: lower mortality but wider spread with Omicron despite enhanced infection-prevention measures. *Infection, Disease & Health*, 28, 81–87. doi:10.1016/j.idh.2022.09.003

[9] Chun J.Y., Jun J.Y., Choi J., et al. (2021). Coronavirus disease 2019 outbreak in a psychiatric closed ward: what we have to learn. *Frontiers in Psychiatry*, 11, 579235. doi:10.3389/fpsyt.2020.579235

[10] Rovers J.J.E., van de Linde L.S., Kenters N., et al. (2020). Why psychiatry is different — challenges and difficulties in managing a nosocomial outbreak of coronavirus disease (COVID-19) in hospital care. *Antimicrobial Resistance & Infection Control*, 9, 190. doi:10.1186/s13756-020-00853-z

[11] Kim J., Choi G., Oh J., et al. (2023). Comparative study on two COVID-19 outbreaks at a long-term mental health facility in Korea in 2020 and 2022. *Medicina*, 59, 1170. doi:10.3390/medicina59061170

[12] Doggui R., Ouakki M., Boulais A., et al. (2026). SARS-CoV-2 outbreaks in long-term care facilities during the Omicron era in Québec, Canada. *Scientific Reports*, 16, 3122. doi:10.1038/s41598-025-32967-y

[13] Espenhain L., Funk T., Kunøe A., et al. (2023). Findings in Danish long-term care facilities in the first year of the SARS-CoV-2 pandemic. *European Geriatric Medicine*, 14, 527–535. doi:10.1007/s41999-023-00793-y

[14] Hospital Authority (2022). *HA Statistical Report 2021–2022* (cluster extracts; service capacity as at 31 March 2022; inpatient discharges and patient-days). Available from: `https://www3.ha.org.hk/data/HAStatistics/StatisticalReport/2021-2022`

[15] Census and Statistics Department, HKSAR (2022). *2021 Population Census: Summary Results* (district populations and densities).

[16] Government of the HKSAR (2017). LegCo Question 13: Resource allocation of Hospital Authority (cluster catchment population projections). `https://www.info.gov.hk/gia/general/201712/13/P2017121300666.htm`